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FisherWright.jl: forward-time Wright–Fisher simulation of whole-genome founder populations in Julia

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Software

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Summary

Stochastic simulations of breeding programs and complex traits require founder populations whose allele frequencies and linkage disequilibrium (LD) determine marker–QTL architecture, genomic prediction accuracy, and long-term selection response. `FisherWright.jl` generates such founders by forward-time simulation of a finite, diploid Wright–Fisher population. The model assumes a constant census size N, two sexes, random union of a sire and dam sampled with replacement, Poisson-distributed crossovers with independent assortment among chromosomes, and neutral mutation under a biallelic finite-sites model. Starting from monomorphic genomes, the simulation runs for a specified number of generations, optionally moves fixed mutations out of the active haplotypes, and exports segregating variants. Output formats include a bit-packed `BitMatrix` with a linkage map, subsampled SNP-chip panels, and native `BnGStructs.jl` haplotype containers designed for downstream breeding simulations. `FisherWright.jl` is implemented in pure Julia (Bezanson et al., 2017), MIT-licensed, and available through the Julia General Registry.

Statement of need

Simulations of genomic selection require founder genomes with site frequency spectra (SFS) and LD profiles that arise from an explicit evolutionary process. Naive Bernoulli sampling produces genotypes in linkage equilibrium, whereas heuristic approximations, such as targeting Sved's expectation $E[r^2] = 1/(1+4N_e c)$, do not reproduce the LD decay of a finite population after minor allele frequency (MAF) filtering. Coalescent simulators generate neutral equilibrium variation efficiently, but their integration into Julia workflows typically requires inter-process communication, foreign-function wrappers, or file exchange with external C and Python executables.

`FisherWright.jl` addresses this gap with an in-process, pure-Julia founder engine. It eliminates cross-language dependencies, returns the data types used by downstream breeding pipelines, and provides a transparent forward-simulation loop that can be inspected, extended, and taught without external tooling.

State of the field

Existing genomic simulation software spans several paradigms. Coalescent engines such as `msprime` (Baumdicker et al., 2022; Kelleher et al., 2016) and `MaCS` (G. K. Chen et al., 2009) are the fastest means of generating neutral equilibrium genomes; `MaCS`, for example, is the default founder generator in `AlphaSimR` (Gaynor et al., 2021). General forward-time simulators such as `SLiM` (Haller et al., 2019, 2026) and `fwdpy11` (Thornton, 2019) accommodate arbitrary selection, complex demography, and spatial structure via the Eidos scripting language or

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Python/C++ interfaces. Specialized breeding simulators, including `QMSim` (Sargolzaei & Schenkel, 2009), `MoBPS` (Pook et al., 2020), and `XSim` (C. J. Chen et al., 2022; Cheng et al., 2015), focus on pedigree management, mating designs, and multi-generation selection programs. Among these, `XSim` version 2 is also implemented in Julia.

`FisherWright.jl` occupies a distinct niche through its deliberately limited scope. It does not implement selection, population structure, or non-stationary demography; workflows requiring those features remain better served by `SLiM` or `fwdpy11`. For neutral equilibrium founders, `msprime` is substantially faster when a Julia-only workflow is not needed. We built a new package rather than contribute to these tools for three reasons. First, the `JuliaBnG` packages needed a founder generator without non-Julia dependencies that produces the `BnGStructs.jl` haplotype type used throughout the pipeline. Second, it needed to be small enough to validate thoroughly against population-genetic theory. Third, its core algorithm originated on 5 December 2010 as a standalone C++ Wright–Fisher simulator (`pgsnp.cpp` in `pop-simulation`), which already represented haplotypes as sorted mutation coordinates, spliced parental strands across sorted crossover breakpoints, and periodically combed out fixed alleles. That code was later ported to Julia inside the author's livestock breeding simulator `xyBnG.jl` (Yu et al., 2026b), which is not publicly available, and then extracted into `FisherWright.jl` so that the founder step could be released, tested, and cited independently.

Software design

Haplotypes are represented as sorted `Vector{UInt32}` coordinates of derived alleles on a concatenated genome of up to $2^{32} - 1$ bp. This sparse representation avoids the prohibitive memory footprint of dense genotype arrays at gigabase scale. Tree-sequence recording would improve the efficiency of forward simulation, but would introduce an additional dependency and a second data model.

Meiosis is implemented as a single two-pointer traversal of the two parental haplotypes, switching strands at each crossover breakpoint (`recombine`). De novo mutations are added by a sorted merge into a preallocated buffer (`merge_sorted!`). These kernels and the crossover sampler (`cobp!`) write to preallocated thread-local buffers; regression tests verify that they perform no heap allocations after warm-up. Within each generation, meioses and mutation draws run in parallel using `Threads.@threads`. The simulation accepts an explicit random-number generator and seeds one `Xoshiro` stream per work slot before entering the threaded loops; mating is sampled serially from the supplied generator. Consequently, a realization is reproducible from the generator state independently of thread count, fixation-extraction interval, and whether fixed substitutions are returned separately. This guarantee holds within a fixed Julia version and package environment: variate algorithms can change between releases (for example, `randn` differs between Julia 1.10 and 1.13, which alters Poisson draws with mean ≥ 6 in Distributions.jl), so the same seed can yield a different realization under another Julia version.

Under neutrality, fixed derived alleles accumulate at the substitution rate μL per generation across L sites and, if retained, progressively reduce meiosis throughput. When tracking is enabled (`result = true`), a sweep at a user-defined interval (every generation by default) removes positions fixed across all $2N$ haplotypes and records them in the substitution log of a `FisherWrightResult` struct. The active haplotype vectors then contain only segregating sites, which bounds memory consumption and traversal overhead during long burn-in periods.

Data export is decoupled from forward simulation. `muts2bitarray` converts polymorphic sites to a 1-bit-per-allele `BitMatrix` with an accompanying `DataFrame` linkage map. For large genomes, where dense matrices are unwieldy, `extract_chip_bitarray` and `to_haplotype(result, positions)` evaluate a sorted subset of marker coordinates directly against sparse haplotypes to generate target SNP panels without instantiating intermediate whole-genome arrays.

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Validation

The validation suite compares `FisherWright.jl` (v0.4.0) with analytical neutral expectations and matched simulations from `msprime`'s discrete-time Wright–Fisher (DTWF) model. The validation setting models $N = 100$ individuals across 10 chromosomes of 10 Mb each ($L = 100$ Mb), evolving for $20N = 2{,}000$ generations at $\mu = r = 10^{-8}$ per base pair per generation, with 50 independent replicates per simulator.

The suite comprises 24 prespecified statistical checks against theory, `msprime`, or both: crossover counts per meiosis (the mean and variance-to-mean ratio for each chromosome) and independent assortment; segregating sites (S) against Watterson's θa_n; gene diversity ($\sum 2pq$) against $\theta(1 - 1/n)$; whole-population and sampled ($n = 20$) site frequency spectra; and LD decay (σ_d^2, MAF ≥ 0.1) across five distance bins from 10 kb to 3 Mb. A check fails when $|z| > 3.5$, where z is calculated from means over the 50 replicates. All 24 checks passed in each of two independent runs (seeds 7 and 2026; maximum $|z|$ values of 2.24 and 2.49, respectively).

A lightweight statistical suite runs in continuous integration for every commit. It passes for five seeds and fails when the simulated N is halved, the recombination rate is doubled, or an earlier bug in final-generation mutation retention (inherited from the 2010 C++ loop and fixed in v0.3.5) is reintroduced. The online documentation includes the complete validation scripts, raw output logs, and a comparison with release v0.1.6.

Research impact statement

`FisherWright.jl` is the founder-population generator for the `JuliaBnG` ecosystem and directly produces `BnGStructs.jl` haplotype containers. An earlier release (v0.1.x) simulated the historical base population for an evaluation of enteric methane breeding objectives in the Norwegian White Sheep program (Yu et al., 2026a). That release predates the corrections listed in the package changelog; their effects on the founders are documented in the manual.

The package repository includes reproducible benchmark drivers. In one run with $N = 2{,}000$ individuals evolved for 2,000 generations ($t = N$, before equilibrium) on a 1 Gb genome (10×100 Mb chromosomes), forward simulation required 59.8 s on 8 threads of an AMD Ryzen 9 3900X. Converting the resulting 567,975 segregating sites across 4,000 haplotypes to a 271 MiB `BitMatrix` required a further 9.0 s. By comparison, `SLiM` with tree-sequence recording required 51.6 s for forward simulation and 11.2 s for variant export, whereas `msprime` using its DTWF model completed the equivalent steps in 9.6 s and 7.6 s, respectively.

AI usage disclosure

Generative AI tools were used during development and writing:

- GitHub Copilot, for code suggestions and for commits after v0.1.6; some of these list Copilot as co-author in the Git history, which was corrected in later commits;
- GPT-5, for an early code review;
- Gemini, for drafts of earlier statistical tests, which were later replaced, and for manuscript editing;
- Google Antigravity, for manuscript wording and code review;
- Claude (Anthropic), for code and manuscript review, validation and comparison scripts, documentation, and a first draft of this paper.

The author reviewed, edited, and validated all AI-assisted output and made the design decisions. Every reported number is drawn from logged runs in the repository.

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